BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: DIK-SSC-IX-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-IX-2022 (A)

Q1. Choose the correct option:

Q1. In NH_3 molecules nitrogen has unshared pair of electron is _____

(A) Four
(B) Three
(C) Two
(D) One

Q2. Which of the following molecule have no hydrogen bonding?

(A) H_2S
(B) H_2O
(C) NH_3
(D) HF

Q3. Molarity is the number of moles of solute present in _____

(A) 1 kg solvent
(B) 1 kg of solution
(C) 1 dm^3 of solution
(D) 1dm^3 of solvent

Q4. According to modern periodic law physical and chemical properties of an elements are the periodic function of their _____

(A) Mass No,s
(B) Atomic No,s
(C) Both A and B
(D) Neutron No,s

Q5. S-block elements are_____

(A) Metals
(B) Non-metals
(C) Metalloids
(D) Transition

Q6. Which of the following is not an examples of amorphuous solid?

(A) Rubber
(B) Glass
(C) Glucose
(D) Plastic

Q7. Shielding effects is due to _____.

(A) Neutron
(B) Proton
(C) Proton and neutron
(D) Electron

Q8. An atom with a charge is called _____

(A) An electron
(B) A molecule
(C) A metal
(D) An ion

Q9. In oxidation _____

(A) An electron is lost
(B) An electron is gained
(C) Hydrogen is gained
(D) Oxygen is lost

Q10. The oxidation number of 'C' in Ch_3 OH is _____.

(A) –2
(B) –1
(C) 0
(D) +1

Q11. What will be the volume of a gas, if the volume of the gas at 2-atmosphere is increased from 1.5dm^3 to 3dm^3?

(A) 1 atmosphere
(B) 1.5 atmosphere
(C) 2 atmosphere
(D) 2.5 atmosphere

Q12. L-shell has sub-shell_____

(A) S
(B) S&P
(C) S,P & d
(D) S,P,d & f

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 53 | **Paper Code:** DIK-SSC-IX-2022 (A)

Q2. Attempt any 8 questions of the following:

(i) What is the mass of 5 moles of ice? Atomic mass of H=1amu O=16amu

(ii) Give the reason that why dative bond is always polar?

(iii) What do you know about the importance of isotopes?

(iv) Differentiate between hard and soft metals.

(v) Define and explain Bowel's Law.

(vi) Why gas is compressible but solid is not compressible? Give reasons.

(vii) How many moles of hydrogen are there in $8.9x10^{23}$ hydrogen atoms?

(viii) Why ionic compounds are good conductors? Compare to covalent compounds.

(ix) How will you differentiate between representative and transition elements?.

(x) Why the densities of a gases is lower than that of liquids?

SECTION – C

Maximum Marks: 21

> Note: Attempt any Three Questions. All questions carry equal marks:

Q3.

(a) What is a Galvanic (voltaic) cell? Briefly describe the cell reaction of Galvanic cells.

(b) Discuss briefly any three techniques of preventing corrosion.

Q4. Explain the following terms.

(a) Electron affinity

(b) Empirical formula and molecular formula of a compound.

Q5.

(a) Write the chemical properties of magnesium and calcium.

(b) What is oxidation and reduction?

Q6.

(a) Draw the Lewis structure for the following, (i). ccl_4 (ii). BF_3

(b) Write down factors which affect vapour pressure.